Table 3 | Examples of expert-annotated principles in four domains and sampled tasks.

Domain	Task	Principles
Creative Non-fiction Writing	Content Writing	The content should be created to encourage readers to engage in interactions, comments, etc.
	Polishing & Editing	The revised content should align with the original text.
	Brainstorming	The content should refrain from pre-judging ideas.
Technical Writing	Content Writing	The generated content should avoid bias toward certain genders, professions, regions, etc.
	Style Transferring	The style of the content should be consistent with the language style specified in the instructions.
Fiction	Content Writing	The perspective should remain consistent with the outline or previous content.
	Outlining	The global outline should not be too brief or general, omitting key plot points.
Marketing Writng	Content Writing	The content of the market writing should be accurate.
	Summarizing	The summarized content should be all-encompassing, leaving out no crucial points.

reward model training and does direct preference optimization.

Specifically, we first invite human experts including professional writers, editors, and content creators to annotate principles for different writing tasks. Different from previous "principle-based" approaches that only write a short description of the principles, for each principle we also collect one case adhering to the principle and one case violating the principle, as well as natural language rationales explaining why the cases adhere or violate the principle. Then we sample a subset of the instruction data with the highest scores in the aforementioned data filtering process and consider them as samples from the optimal policy as the output texts are carefully selected and instruction-output pairs are top-ranked. For each sample, we first present the principles for the task and ask GPT to analyze which principle can best explain why the response is of good quality. We then ask GPT to synthesize a counterpart of the response violating the principle while adding minimal modifications and do not affect other good aspects of the original response.

With the collected data, we consider the original-perturbed response pairs as (y_w, y_l) pairs and do standard DPO training. In this way, each data pair contains critical training signals about the corresponding principles and helps fine-tune the model to follow the principles. The preference data synthesized by our approach contains much less noise compared to standard RLAIF pipeline, especially in writing domains since LLMs struggles to do literary critic. Compared to RLCD, the most related method for preference data generation, we consider high-quality SFT data instead of LLM-generated as positive examples and use expert-written principles for negative example generation. This makes the training signal less noisy and more principled.

4. Alignment

4.1. Supervised Fine-tuning

4.1.1. Data

To collect the dataset for supervised fine-tuning, we first collect high-quality content written by human writers and content creators according to their metadata including their ratings, number of reads, upvotes, and comments. We adopt the aforementioned data synthesis framework to synthesize instruction following data covering 30+ fine-grained domains and over 10 tasks, instruction annotation data, text generation evaluation data, retrieval-augmented generation data, and function calling data. The combined instruction tuning dataset consists of around 1,000,000 samples. We then run the data filtering process and select 400,000 data points as the final dataset for supervised fine-tuning.

4.1.2. Training

We fine-tune the continual pre-trained models for 3 to 5 epochs. We use a cosine learning rate scheduler with a peak learning rate of 1e-5 and 2e-5 for larger models (i.e., Weaver Ultra and Weaver Pro) and 4e-5 for smaller models (i.e., Weaver Base and Weaver Mini) with 5% warmup steps. We train all models with a global batch size of 256. After supervised fine-tuning, we select the best-performing checkpoint on an internal validation set for preference optimization.

4.2. Preference Optimization

4.2.1. Data

For preference optimization, we select 500 highest-rated samples in the data filtering stage for each subdomain as positive examples for the Constitutional DPO pipeline. We collect over 200 principles and their corresponding few-shot exemplars. We generate one negative example per positive example, resulting in 25,000 preference data pairs.

4.2.2. Training

We fine-tune the supervised fine-tuned models using the conventional DPO algorithm. We train our models for three to five epochs. We use a linear learning rate scheduler with a peak learning rate of 5e-7 and 5% warmup steps. We train Weaver Ultra using a global batch size of 40, while for the others we use 32 and set $\beta = 0.1$. We select the best-performing checkpoint on the internal validation set as the final Weaver models.

5. Evaluation

5.1. WriteBench

Most existing benchmarks for LLMs (Zheng et al., 2023a) and natural language generation (Jiang et al., 2023c; Lin et al., 2020) focus on the reasoning ability or the general-purpose instruction following ability instead of the ability of LLMs to produce creative, stylish, and human-like text content. To this end, we construct WriteBench, a new benchmark for assessing the writing capabilities of LLMs[3].

Similar to how we collect training data for Weaver, WriteBench is designed to cover multiple domains and tasks. To ensure a fair comparison between Weaver and compared generalist LLMs, the

[3]WriteBench will be publically available at `https://github.com/aiwaves-cn/WriteBench`

data collection and data selection process for instructions in WriteBench is done by our independent evaluation team. The resulting WriteBench consists of over 1000 testing instructions covering four domains including fiction writing, creative non-fiction writing, technical writing, and marketing writing. The first release of the WriteBench benchmark is in Chinese since we want to measure the Chinese writing capabilities of the compared models.

5.2. Compared Models

We compare Weaver with competitive Chinese LLMs including both open-sourced models and proprietary models of different sizes, including GPT-4, GPT-3.5, GLM-4, Claude2, Gemini Pro, ERNIE-Bot-4.0, ERNIE-Bot-3.5, Qwen-72B-Chat, Qwen-14B-Chat, Qwen-7B-Chat, Qwen-1.8B-Chat, YI-34B-Chat, YI-6B-Chat, and ChatGLM3-6B. We directly use the same instructions in WriteBench as input prompts for all tested LLMs and collect the model outputs as responses.

Table 4 | LLM-based Evaluation Results

Models	Style	Relevance	Creativity	Overall
Weaver Ultra	8.94	8.96	7.71	8.54
GLM-4	8.83	9.55	6.58	8.32
GPT-4	8.80	9.45	6.32	8.19
Weaver Pro	8.52	8.45	7.3	8.09
YI-34B-Chat	8.70	9.17	6.26	8.04
Claude2	8.42	8.89	6.41	7.91
Qwen-72B-Chat	8.47	8.98	5.95	7.80
Weaver Base	8.61	8.81	5.89	7.77
Qwen-14B-Chat	8.51	8.85	5.89	7.75
Weaver Mini	8.41	8.38	6.35	7.71
Gemini Pro	8.39	8.79	5.88	7.69
Qwen-7B-Chat	8.40	8.80	5.81	7.67
Yi-6B-Chat	8.24	8.67	6.00	7.64
ChatGLM3-6B	8.16	8.70	5.86	7.57
GPT-3.5	8.37	8.65	5.60	7.54
ERNIE-Bot-3.5	8.24	8.22	5.71	7.39
ERNIE-Bot-4.0	8.15	8.05	5.61	7.27
Qwen-1.8B-Chat	7.97	7.86	5.66	7.16

5.3. LLM-based Evaluation

We first perform an LLM-based evaluation to do a coarse-grained evaluation of the compared models. We use GPT-4 as the judge to score each instruction-response pair following the practice and prompt templates in MT-Bench. The results are shown in Table 4. We find that in terms of writing style and creativity, Weaver Ultra significantly outperforms all proprietary models including strong competitors such as GPT-4 and GLM-4. GPT-4 and GLM-4 are better at the relevance metric because they are at least few times larger than Weaver Ultra and thus have better instruction-following ability. As for Weaver of other sizes, we can see that with only 14B parameters, Weaver Pro outperforms all open-source models including those with 70B and 34B parameters, as well as most proprietary models. Similarly, Weaver Base and Weaver Mini are also comparable with generalist LLMs with more than two times their sizes. Overall, the results confirm the effectiveness of our data synthesis and training framework for LLMs specialized in creative writing.